Personal Details

Dominic Bräm
Pediatric Clinical Pharmacology and Pharmacometrics
University Children's Hospital Basel

Education & Work Experience

2025 – present — **Postdoctoral researcher in Pediatric Clinical Pharmacology and Pharmacometrics, University Children's Hospital Basel UKBB**
Developed machine learning algorithms for automatic pharmacometric model development, applied model-informed drug dosing approaches to improve clinical care in pediatrics

2026 — **1-Month research exchange at Uppsala University, Faculty of Pharmacometrics**
Supported the implementation of neural ordinary differential equations to model pharmacometric data in leishmaniasis as part of a research project at Uppsala University

2021 – 2024 — **PhD student in Pediatric Clinical Pharmacology and Pharmacometrics, University of Basel (graduated with *Summa cum laude*)**
Developed neural ODE methods and computational workflows for pharmacometric model development, dose optimization, covariate imputation, and pharmacokinetic analysis.

2020 – 2021 — **Internship on Advanced Analytics in Pharmacometrics at Roche**
Investigated the application of artificial neural networks in PKPD modelling

2019 – 2020 — **Master Thesis: *Population Pharmacokinetics in Patient Data and Image Analysis***
Investigated population pharmacokinetics in pediatrics for Busulfan and Gentamycin, analyzed PET-data using ordinary differential equations and analytical approaches

2019 — **Research Project: *Interrupted time series analysis of acetaminophen intoxications and the influence of a policy change***
Statistical time series analysis of spontaneous case reports, investigated effect of new dosage strength on acetaminophen intoxications

2018 – 2020 — **MSc Pharmaceutical Sciences, ETH Zurich**
Courses in Biostatistics, Pharmacoepidemiology, Computer-Assisted Drug Design, Scientific writing, Pharmacovigilance, Clinical Development etc.

2015 – 2018 — **BSc Pharmaceutical Sciences, ETH Zurich**
General courses in Mathematics, Statistics, Informatics, Biology, Chemistry, Biochemistry etc.

Teaching Experience

2024 – present — **Teaching: Informatics of pharmacometrics at University of Applied Sciences and Arts Northwestern Switzerland FHNW**
Design and deliver an introductory pharmacometrics course on Master's level.

2022 – present — **Teaching: R introduction course at University of Basel**
Design and teach an introductory R course for students and staff, combining theoretical instruction with hands-on exercises.

Workshops

2026 — **Invited Workshop: Applying pmxNODE to Neural ODE-Based Pharmacometric Modeling, Chinese University of Hong Kong**
Developed and delivered a hands-on workshop on implementing neural ODE models using the R package pmxNODE.

2026 — **Invited workshop: Neural ODEs for Automated Pharmacometric Model Development at Uppsala University**
Led a workshop on using neural ODEs to support automated model development in pharmacometrics.

2025 — **Invited workshop: Neural Ordinary Differential Equations in Pharmacometrics at University of Copenhagen**
Delivered theoretical and hands-on training for academic and industry participants on fitting pharmacometric data and supporting automated model development with neural ODEs in Monolix.

2023 — **Invited workshop: Machine Learning in Pharmacometrics at University at Buffalo**
Developed and delivered theoretical and hands-on training on integrating machine learning with pharmacometric modeling in Monolix.

2023 — **Workshop: OptiDose for Individualized Dose Optimization in NONMEM at the PAGE conference**
Delivered a hands-on workshop on applying the OptiDose algorithm to estimate individualized dosing regimens.

Supervising and Mentoring

2026	**Master's Thesis Supervisor** Implementation of a Python-Based Framework for Pharmacometric Modeling
2026	**Supervisor, External PhD-Level Industry Internship** Supervised a PhD researcher evaluating pmxNODE using pharmacometric data from clinical drug-development studies at Roche.
2025	**Supervisor, Two Master's Research Projects** Supervised two FHNW students conducting pharmacometric simulations and analyses of caffeine pharmacokinetics in neonates at UKBB.
2022 – 2023	**Mentor, Applied Pharmacometrics Training Fellowship** Mentored four PhD-level scientists from Africa in pharmacometric modeling and dose analysis of anthelmintic therapies, in collaboration with the Swiss Tropical and Public Health Institute (TPH).

Selected Invited Talks and Conference Presentations

2026	**Learning from Machine Learning in Pharmacometric Modelling** EMBL-EBI Next-Generation ADMET modelling for drug discovery, Hinxton, UK
2025	**Pharmacometrics Modeling to Inform Dose Optimization** University of Cambridge and Astra Zeneca Clinical Pharmacology training, Cambridge, UK
2024	**Modelling pharmacometric data with Neural ODEs** DMDG Open Meeting, York, UK
2024	**Modeling pharmacometric data with Neural ODEs: Get insights into unknown model structures with machine learning** Basel Model & Simulation Seminar, Basel, CH
2023	**Modelling pharmacometric data with Neural ODEs including inter-individual variability** American Conference on Pharmacometrics (ACoP), Maryland, USA
2023	**Neural ODE structures based on pharmacokinetic principles** Population Approach Group Europe (PAGE) conference, A Coruña, ESP
2022	**Machine Learning supports covariate imputation in Pharmacometrics** American Society for Clinical Pharmacology & Therapeutics (ASCPT) Satellite Session
2021	**Supporting Pharmacometrics by learning with machines** Basel Modeling & Simulation Seminar, Basel, CH

Research Software

1. **pmxNODE:** Open-source R package for integrating low-dimensional neural ordinary differential equation models with Monolix, NONMEM, and nlmixr2.https://cran.r-project.org/web/packages/pmxNODE/index.html

2. **optiDoseR:** Open-source R implementation of the OptiDose algorithm for estimating individualized dosing regimens in pharmacometric software. https://github.com/braemd/optiDoseR

Patents

1. **Bräm, DS**, Koch, G, Pfister, M. Computer system and method for generating a data driven interpretable pharmacometrics prediction model to predict medical state parameters of a patient of a special population. Submitted to European Patent Office 26.04.2024 [Published]

2. **Bräm, DS**, Parrot, N, Hutchinson, L, Steiert, B. Prediction of pharmacokinetic curves. Submitted to European Patent Office, filing date 14.07.2022, publication date 19.01.2023, File number EP4371118 [Decision pending]

Certificates and Awards

2024	**PhD graduated with *Summa cum laude*, University of Basel**
2022	**Prize for best poster presentation, UKBB Research Day**
2020	**GSIA Prize for Master's thesis, ETH Zurich**
2018	**Bachelor graduated within the top 5% of students, ETH Zurich**

Extracurricular Activities

2010 - present	**Scout leader, Group leader, Section leader, Section president** Lead an association of more than 120 members and coordinate a team of up to 20 scout leaders, with responsibility for annual planning, events, governance, and organizational communications.
2018 - 2020	**Editor-in-Chief, Tonikum Student Magazine** Coordinated editorial meetings and article submissions and managed sponsor communication, printing, and publication.
2019	**SPHN Data Privacy and IT Security Training** Training for the handling of patient data

Publication list (https://orcid.org/0000-0001-9094-8361)

1. **Bräm, DS**, Steiert, B, Steffens, B, Pfister, M, Koch, G. Automated Pharmacometric Model Development by Leveraging Low-Dimensional Neural ODEs and LASSO Regression. CPT: Pharmacometrics & Systems Pharmacology. 2026. doi: 10.1002/psp4.70285

2. **Bräm, DS**, Gotta, V, Koch, G, Golhen, K, Woerner, A, Atkinson, A, van den Anker, J, Pfister, M, Welzel, T. Evaluation of adalimumab tapering regimens in children and adolescents with juvenile idiopathic arthritis in remission: a simulation study. Pediatr Rheumatol. 2026. doi: 10.1186/s12969-026-01184-w

3. **Bräm, DS**, Gotta, V, Welzel, T. Therapieintensivierung in der Rheumatologie am Beispiel der nicht-systemischen juvenilen idiopathischen Arthritis: die Sicht der Pharmakologie. Rheuma plus. 2025. doi: 10.1007/s12688-025-00862-z

4. **Bräm, DS**, Bachmann, F, Schropp, J, Gotta, V, Steffens, B, van den Anker, J, Pfister, M, Koch, G. Understanding and Streamlining Dose Finding: From Dose Simulation to Dose Estimation. J Clin Pharm. 2025 65: 731-741. doi: 10.1002/jcph.6188

5. **Bräm, DS**, Steiert, B, Pfister, M, Steffens, B, Koch, G. Low-dimensional neural ordinary differential equations accounting for inter-individual variability implemented in Monolix and NONMEM. CPT Pharmacometrics Syst Pharmacol. 2025; 14: 5-16. doi: 10.1002/psp4.13265

6. **Bräm, DS**, Koch, G, Allegaert, K, van den Anker, J, Pfister, M. Applying Neural ODEs to Derive a Mechanism-Based Model for Characterizing Maturation-Related Serum Creatinine Dynamics in Preterm Newborns. J Clin Pharm. 2024; doi: 10.1002/jcph.2460

7. **Bräm, DS,** Nahum, U, Schropp, J, Pfister, M, Koch, G. Low-dimensional neural ODEs and their application in pharmacokinetics. J Pharmacokinet Pharmacodyn 2023; doi: 10.1007/s10928-023-09886-4

8. **Bräm, DS**, Parrott, N, Hutchinson, L, Steiert, B. Introduction of an artificial neural network–based method for concentration-time predictions. CPT Pharmacometrics Syst Pharmacol. 2022; 11: 745– 754. doi: 10.1002/psp4.12786

9. **Bräm, DS,** Nahum, U, Atkinson, A, Koch, G, Pfister, M. Evaluation of machine learning methods for covariate data imputation in pharmacometrics. CPT Pharmacometrics Syst Pharmacol. 2022; 11: 1638- 1648. doi: 10.1002/psp4.12874

10. Rohleff, J, Bachmann, F, Nahum, U, **Bräm, DS**, Steffens, B, Pfister, M, Koch, G, Schropp, J, Redefining Parameter Estimation and Covariate Selection via Variational Autoencoders: One Run Is All You Need. CPT Pharmacometrics Syst Pharmacol. 2025, 14: 2232-2243; doi: 10.1002/psp4.70129

11. Ponce-Bobadilla, AV, **Bräm, DS**, et al. Predictive AI in Clinical Pharmacology: A Call to Action to Develop Robust Benchmarking Practices. CPT Pharmacometrics Syst Pharmacol. 2025; 15: no.1. doi: 10.1002/psp4.70155.

12. Golhen, K, Welzel, T, Atkinson, A, Koch, G, **Bräm, DS**, van den Anker, J, Woerner, A, Pfister, M and Gotta, V. Influence of Disease Type and Activity on Adalimumab Exposure in Children with Inflammatory Rheumatic Diseases. J Clin Pharm. 2025. doi: 10.1002/jcph.70045

13. Ajayi, DT, Orherhe, OM, Pillai, G, Mouksassi, S, Steffens, B, **Bräm, DS**, Sprecher, V, Hofmann, D, Buettcher, M, Coulibaly, J, Ali, SM, Keiser, J, Pfister, M. Pharmacometric Analysis to Describe

Pharmacokinetics and Exposure-Efficacy Response of Ivermectin in Adolescents Infected with Trichuris trichiura. J Clin Pharm. 2025 65: 433-440. doi: 10.1002/jcph.6158

14. Gotta, V, Bielicki, JA, Paioni, P, Csajka, C, **Bräm, DS**, Berger, C, Giger, E, Buettcher, M, Posfay-Barbe, KM, Van den Anker, J, Pfister, M. Pharmacometric in silico studies used to facilitate a national dose standardisation process in neonatology - application to amikacin. Swiss Med Wkly. 2024 8;154:3632. doi: 10.57187/s.3632.

15. Pillay-Fuentes Lorente, V, Nwogu-Attah, J, Steffens, B, **Bräm, DS**, Sprecher, V, Hofmann, D, Buettcher, M, Pillai, G, Mouksassi, S, Coulibaly, J, Pfister, M, Keiser J. Understanding Drug Exposure and Trichuris trichiura Cure Rates: A Pharmacometric Approach for Albendazole-Ivermectin Co-medication in Tanzania and Côte d'Ivoire. Drugs R D. 2024. doi: 10.1007/s40268-024-00476-4

16. Schreib KM, **Bräm DS**, Zeilhofer UB, Müller D, Güngör T, Krämer SD, Hauri-Hohl MM. Population Pharmacokinetic Modeling for Twice-Daily Intravenous Busulfan in a Large Cohort of Pediatric Patients Undergoing Hematopoietic Stem Cell Transplantation-A 10-Year Single-Center Experience. Pharmaceutics. 2023 Dec 20;16(1):13. doi: 10.3390/pharmaceutics16010013. PMID: 38276491.

17. Ribba, B, **Bräm, DS**, Baverel, PG, Peck, RW. Model enhanced reinforcement learning to enable precision dosing: A theoretical case study with dosing of propofol. CPT Pharmacometrics Syst Pharmacol. 2022; 11: 1497- 1510. doi:10.1002/psp4.12858

18. Husmann, F, Stierli, L, **Bräm, DS,** et al. Kinetics of iron absorption from ferrous fumarate with and without galacto-oligosaccharides determined from stable isotope appearance curves in women, The American Journal of Clinical Nutrition, 2022, Volume 115, Issue 3, Pages 949–957, doi:10.1093/ajcn/nqab361

19. Paioni P, Jäggi VF, Tilen R, **Bräm, DS,** et al. Gentamicin Population Pharmacokinetics in Pediatric Patients—A Prospective Study with Data Analysis Using the saemix Package in R. Pharmaceutics. 2021; 13(10):1596. Doi:10.3390/pharmaceutics13101596

20. Martinez-De la Torre A, Weiler S, **Bräm DS,** Allemann SS, Kupferschmidt H, Burden AM. National Poison Center Calls Before vs After Availability of High-Dose Acetaminophen (Paracetamol) Tablets in Switzerland. JAMA Netw Open. 2020;3(10):e2022897. doi:10.1001/jamanetworkopen.2020.22897

21. Mu, L, Krämer, SD, Warnock, GI, **Bräm, DS**, et al. [11C]mHED PET follows a two-tissue compartment model in mouse myocardium with norepinephrine transporter (NET)-dependent uptake, while [18F]LMI1195 uptake is NET-independent. 2020, EJNMMI Res 10, 114. doi:10.1186/s13550-020-00700-7